QuanTiMedAI: Quantum-Enhanced Time-Series AI for Cardiac Arrest Mortality Prediction

Agentic LLM-Guided Quantum Deep Learning for ICU Decision Support in Resource-Limited Hospitals

ABSTRACT

Cardiac arrest mortality ICU patients can reach 82%, yet existing models rely on static admission baselines and ignore how fast the patients condition is changing. We developed QuanTiMedAI, which analyzes dynamic ICU vitals and lab data into continuous time series, using an agentic LLM to pick clinically meaningful features, feeding into a Quantum LSTM architecture. Our framework achieved an AUROC of 0.852 on MIMIC-IV, the best result reported for this task by using only 605 parameters. In resource-limited settings like Bangladesh, where ICU workflows are mostly manual, this architecture enables real-time bedside mortality forecasting without demanding high performance computing infrastructure.

Method with System Diagram/Design Complexity

QuanTiMedAI has a two-stage process. First, an agentic LLM (Gemma) reviews training-set statistics and selects the most clinically meaningful vitals and labs, iteratively refining its choices with cross-validation feedback rather than simple correlation filtering. Next, these features are converted into hourly time-series blocks and fed into a custom five-circuit Quantum LSTM (QLSTM), which replaces classical gating mechanisms with variational quantum circuits and includes a residual connection to maintain original signal intensity. The final hidden state is fed into a small feedforward classifier that outputs mortality probability as shown in the architecture diagram below.

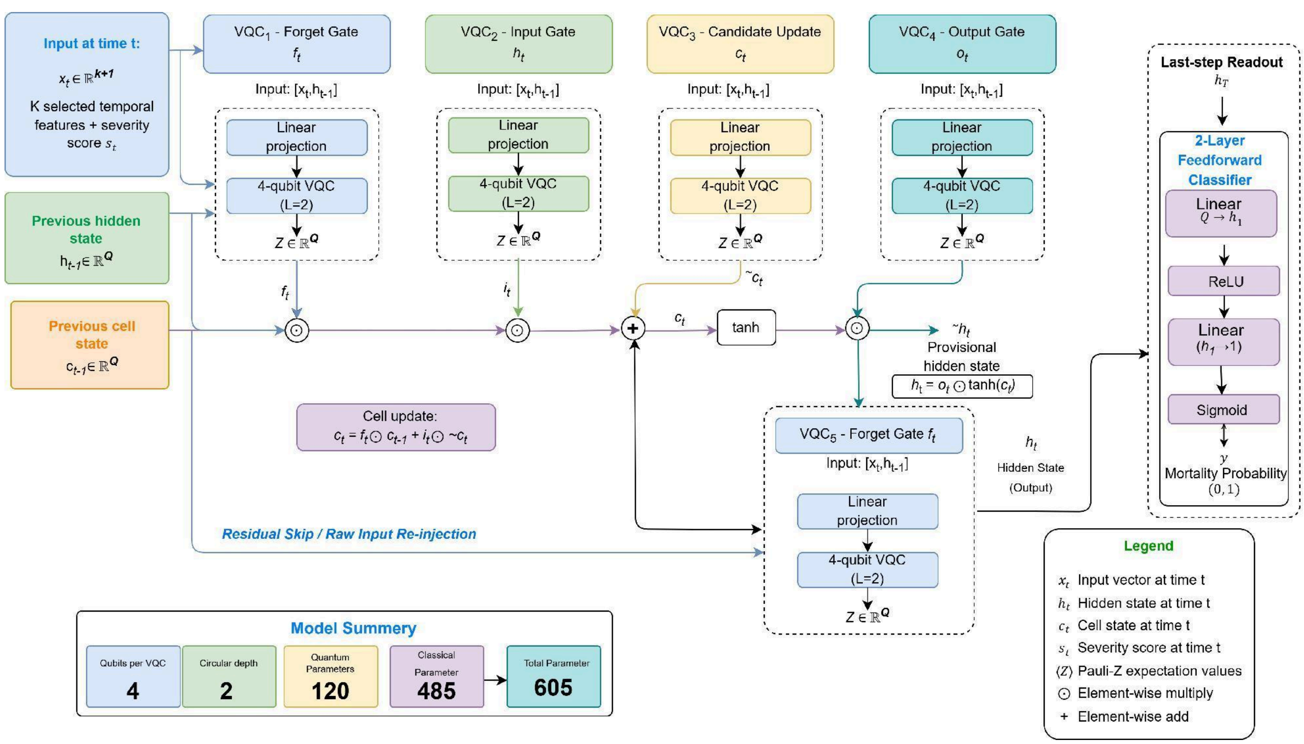

Figure 1: Proposed five-VQC QLSTM cell with residual input re-injection, showing the forget, input, candidate, and output gates feeding the last-step readout and classifier head

Novelty of Project

This work is the first MIMIC-IV cardiac arrest framework using time-series modelling paired with clinician-like LLM feature selection. We are also the first to apply Quantum LSTM to clinical EHR data, outperforming the best classical baseline (0.852 vs 0.828 AUROC) with 466 times fewer parameters.

Impact on society/environment

In Bangladesh, ICU assessments remain largely manual, and hospitals often lack the computing resources for heavy AI models. A 605-parameter model can run on standard hardware and give clinicians a real-time mortality prediction, helping triage patients and allocate scarce ICU beds and staff more efficiently in resource-limited settings.

Business Model/Feasibility/Financial Scalability plan

QuantMedAI's lightweight design allows it to be integrated into hospital ICU monitoring systems and run on standard servers, as its quantum circuits are simulated without specialized hardware. We plan to pilot it with partner hospitals in Bangladesh as a low-cost add-on to existing EHR and monitoring systems, then expand through institutional licensing to public and private ICUs across Bangladesh and other resource-limited settings. Revenue would come from subscriptions rather than per-patient fees, making adoption more practical for underfunded hospitals.

Conclusion

QuanTiMedAI is fully developed and evaluated on MIMIC-IV, including the agentic feature selection pipeline, the QLSTM architecture, and an ablation study, achieving state-of-the-art AUROC. The work is under review at IEEE JBHI and available on arXiv.